Response to Reviewer 2

Enhancing Consent Policy Privacy in Blockchain-Based Healthcare Data Sharing

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Dear Reviewer 2,

We thank you for your careful review and constructive comments, which have helped us to improve the manuscript considerably. Our point-by-point responses follow and a marked manuscript in which all new or revised text appears in red is provided.

Comments 1: *The abstract is clear and presents the manuscript; however, it is too long (235 words). It must be less than 200 words.*

Response 1: We rewrote the abstract, which is now shorter than 200 words. It also reports the re-measured padding overhead of the revised system.

Revised in: Abstract (page 1, lines 2–18).

Comments 2: *Some of the keywords are wrong, such as "policy privacy" – 4 times in the text and "data use ontology" – 7 times.*

Response 2: We revised the keywords to match the terms used in the manuscript. "Policy privacy" is now "consent policy privacy" (the phrase in the title), "data use ontology" is now "Data Use Ontology (DUO)", and "zero-knowledge proof" is now "zero-knowledge proofs". We also added "metadata leakage", the central concern of the paper.

Revised in: Keywords (page 1, lines 19–20).

Comments 3: *The introduction correctly outlines the issues. However, rows 123–131 point to old sources and methods of encryption. It raises the question "why?" – do the authors not know the newest or do they not want to use the newest; why do the authors not comment on the newest solutions?*

Response 3: We kept the original CP-ABE and PRE citations because they define the primitives this paragraph describes. We added recent blockchain healthcare designs built on them: PRE-based sharing of medical records and IoMT data [19,20] and policy-hiding CP-ABE, one of them with distributed key generation [13,14]. The Related Work section also discusses recent hidden-policy, self-sovereign-identity, and zero-knowledge authentication work (see Response 10).

Revised in: Section 2.1 (pages 3–4, lines 120–133); Section 3, third and fourth paragraphs (pages 5–6, lines 189–209; page 6, lines 210–232).

Comments 4: *There is so much empty space on page 6, page 20, and page 25 because of the wrong position of Table 2, Table 9, and Table 12.*

Response 4: The gaps were caused by forcing every float to its source position. All tables and figures now use standard float placement, we adjusted the float-page settings, and the abbreviations list can now break across pages. We checked the recompiled manuscript page by page, and the large empty areas are gone.

Revised in: Whole manuscript; the former Tables 2, 9, and 12 are now Tables 2 (page 7), 11

(page 26), and 14 (page 29).

Comments 5: *The titles of tables and figures are very long. For example, Figure 1 has a 4-row title and Table 12 a 6-row title.*

Response 5: We shortened every caption to one or two sentences and moved the explanations into the main text. For example, the caption of Figure 1 now fits on one line (formerly four), and that of the former Table 12 (now Table 14) on three (formerly six).

Revised in: All figure and table captions, for example Figure 1 (page 12) and Table 14 (page 29).

Comments 6: *On row 681 there is a formula in the text. This is not acceptable. Follow the proper format of formulas.*

Response 6: We removed the inline formulas from the results text. Statistics are now given as plain numbers (for example, "mean AUC 0.667 with 53% of seeds significant"), and the full values remain in the tables. The only formula in this part of the evaluation, the paired leakage metric, is now a displayed and numbered equation (Equation (16)).

Revised in: Section 6.4, paragraph "Supervised attack" and the paragraph that follows (pages 22–23, lines 839–849); Section 6.5, last paragraph (page 24, lines 901–910); Section 6.6, paragraph "Supervised attack" and the paragraph that follows (page 25, lines 926–946); Section 6.9, fourth paragraph (pages 28–29, lines 1022–1030).

Comments 7: *The opinion of the authors about the results is very short. The conclusions in Part 6 are almost missing.*

Response 7: We have expanded both parts. The Discussion now opens with our interpretation of the results, including the two criterion failures and why we report them as failures. The Conclusion has been rewritten as four paragraphs: contribution, findings, limitations, and future directions.

Revised in: Section 7, paragraphs under "Privacy findings" (pages 31–32, lines 1114–1136); Section 8 (pages 34–35, lines 1229–1261).

Comments 8: *The English language looks quite good. The title and abstract reflect the content.*

Response 8: The title is unchanged; the abstract was shortened as described in Response 1.

Comments 9: *The discussion and conclusion summarize the proposed idea. However, there is a repetition of the same conclusion in the abstract, discussion, and conclusion. This repetition of the same information should be avoided.*

Response 9: We removed the repetition by giving each part a distinct role. The Abstract states the results and overhead figures once. The Discussion interprets them (the two criterion failures, cost, and trust model) without restating the Abstract. The Conclusion summarizes the contribution, findings, limitations, and future directions without repeating the task counts or overhead figures.

Revised in: Abstract (page 1, lines 2–18); Section 7, paragraphs under "Privacy findings" (pages 31–32, lines 1114–1136); Section 8 (pages 34–35, lines 1229–1261).

Comments 10: *Many of the references are old [7, 8, 10, 14, 15, 16, 17, 18, 19, 21, 23, 24,*

25, 26, 27, 30, 31]. Some of them do not follow the proper format. I offer the authors to find newer sources and replace these old sources.

Response 10: We added seven references from 2024–2026 and cite them where the text relies on them. They cover PRE-based medical data sharing [19,20], policy-hiding CP-ABE [13,14], self-sovereign identity [27], zero-knowledge patient authentication [28], and a survey of zero-knowledge proofs for blockchain privacy [29]. The list now has 41 entries, 12 of them from 2024 or later (previously 31 and 5); the total also includes three foundational references for the Poseidon hash, EdDSA, and the Baby Jubjub curve, which the revised trust model now introduces explicitly. We kept the older references that are original constructions or standards used in the paper (for example Groth16, Zerocash, CP-ABE, PRE, ICD-10, and OBI), since no newer source replaces them. We also corrected the format of the AZTEC, Zeth, and Ibraimi et al. entries and added DOIs.

Revised in: Section 2.1 (pages 3–4, lines 120–133); Section 3, third and fourth paragraphs (pages 5–6, lines 189–209; page 6, lines 210–232); References.